

Local LLM planner

Llama-3.1-8B, served locally

`{"kind": "action", "action_id": "auth_log"}` (3 of 3 decisions)

proposes probes but never issues a verdict; it sees no ranking

propose

finish (never used)

1 Hypothesis Ledger

	prior	IPs	HTTP	admin
credential stuffing	.11	0	0	0
SQL injection	.11	.24	0	0
exposed admin panel	.11	.24	.33	.97
authorized scan	.11	0	0	0
monitoring FP	.11	0	0	0
insider exfiltration	.11	0	0	0
vulnerable component	.11	.24	.33	0
DNS C2	.11	.24	.33	0
other	.11	.05	.02	.02

 $P(o \mid h, a)$ counted from LLM-free runs

Bayes update

2 EIG/cost Probe Selection

step 1 of 3: expected information gain per cost

source_ips  1.74 ← run
access_pattern  1.36
admin_exposure  0.51
auth_log  0.51 ← LLM's pick
change_ticket  0.51

3 probes

Read-only probes

o1: source_ips → normal
o2: access_pattern → normal
o3: admin_exposure → exposed_no_auth

3 Evidence Rule & Stop

leader $p = 0.97 \geq 0.8$ ✓
cites current evidence o3 ✓

Controller stop

Verdict: exposed admin panel
cites o3, o2, o1

Independent verifier ✓



alert
sec-base-02

Journal & Audit

request

decision

prediction

observation

update

×3

stop

verify

each prediction written *before*
its observation; audit replays it ✓